Supersonic and Hypersonic Aerodynamic Extensions for OpenRocket with Experimental Validation from Mach 0.3 to 17

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Summary

This work extends OpenRocket (Niskanen 2009) by replacing the aerodynamic model suite with a physics-based implementation valid from Mach 0.3 through Mach 17+. The original OpenRocket aerodynamic core, built on the Barrowman slender-body method (Barrowman 1967), was designed for subsonic flight and produces increasingly inaccurate results above approximately Mach 0.8. The extended implementation adds validated supersonic and hypersonic models to the same well-structured Java codebase while preserving full backward compatibility with existing `.ork` rocket design files and the familiar six-degree-of-freedom trajectory simulator. These extensions are developed as a fork with the goal of eventual integration into the main OpenRocket project.

The principal additions are: a complete oblique/normal shock solver and Prandtl-Meyer expansion package (Ames Research Staff 1953); a shock geometry pre-pass that threads post-shock local flow conditions to all downstream components; Taylor-Maccoll exact wave drag for conical noses (Taylor and Maccoll 1933); shock-expansion theory for ogive bodies; DATCOM Section 4.1.5.1 fin wave drag (Finck 1978); a four-model base drag suite covering turbulent, laminar, and boattail regimes (Chapman 1950; Engineering Sciences Data Unit 1977); Van Driest II compressible skin friction (Hopkins and Inouye 1971); and Modified Newtonian theory for the hypersonic regime (Anderson 2006). All Mach regime transitions use C1-continuous polynomial blending to prevent trajectory integrator instabilities near Mach 1. The software is validated through 72 aerodynamic test files across 22 independently benchmarked subsystems.

Statement of Need

High-power rocketry (HPR) vehicles, university sounding rockets, and amateur research projects routinely reach Mach 2 to 5 and beyond. Yet the tools available for open-source trajectory simulation — OpenRocket, RocketPy (Ceotto et al. 2021), and OpenTsiolkovsky — all rely on the Barrowman equations, which assume subsonic, shock-free, incompressible-boundary-layer flow. Six specific failures limit the original OpenRocket above approximately Mach 0.8:

1. A hard-clamped compressibility factor (`MIN_BETA = 0.25`) that holds beta constant through the transonic regime instead of following $\beta = \sqrt{|M^2 - 1|}$, producing a flat plateau in every coefficient that depends on $1/\beta$.
2. No wave drag computation — fin drag is skin friction only, causing systematic under-prediction of total drag by 10–20% at Mach 2.
3. Tabulated nose-pressure data limited to approximately Mach 3, with linear extrapolation above that producing physically incorrect results.
4. Linear fits for speed of sound and viscosity that err by 5–50% outside the near-sea-level temperature range.
5. No shock wave geometry — every component receives freestream conditions even though the nose shock reduces local Mach by 10–25% at the fin station.
6. No supersonic center-of-pressure correction — the CP shift of 5–10% of body length that occurs supersonically is absent, with implications for stability margin predictions.

The commercial tool RASAero II (Rogers 2015) provides the accuracy needed for HPR and research work, but is closed-source, Windows-only, and cannot be extended or independently verified. No open-source tool currently combines trajectory simulation with validated supersonic aerodynamics. This work fills that gap.

State of the Field

Table 1 summarizes the aerodynamic capabilities of the principal tools in the field.

Tool	Trajectory	Supersonic Aero	Open Source
OpenRocket (original)	Yes	No (Barrowman, M $<$ 0.8)	Yes
RASAero II	Yes (3-DOF)	Yes (validated)	No
RocketPy	Yes	No (Barrowman)	Yes
Digital DATCOM	No	Yes (empirical methods)	No
OpenFOAM / SU2	No	Yes (CFD)	Yes
This work	**Yes**	**Yes (validated)**	**Yes**

Digital DATCOM (Finck 1978) provides many of the same aerodynamic methods but requires substantial expertise to operate and produces coefficient tables rather than integrated trajectories. Full CFD codes such as OpenFOAM or SU2 can resolve supersonic flows in detail but are impractical for iterative vehicle design or flight simulation. The present work uniquely occupies the combination of trajectory simulation, validated supersonic and hypersonic aerodynamics, and a fully open, extensible codebase.

Software Design

The extended aerodynamic model is written in Java and built within OpenRocket's multi-module Gradle system (core simulation library plus Swing UI). The aerodynamic pipeline follows the original Barrowman pattern — an orchestrating `BarrowmanCalculator` delegates to per-component calculators — extended by a supersonic pre-pass:

```
BarrowmanCalculator
+-- ShockGeometry.compute()       [supersonic pre-pass, inert at M < 1]
+-- BarrowmanStabilityCalculator [CN, CP, Cmq, Magnus, vortex sideforce]
+-- BarrowmanDragCalculator      [friction + pressure + base + power-on]
```

`ShockGeometry` is computed once per timestep. It walks the rocket nose-to-tail, applies the theta-beta-Mach relation and Taylor-Maccoll cone flow at the nose, and marches downstream through body transitions using oblique shock and Prandtl-Meyer expansion relations to produce local Mach, pressure, and temperature at each axial station. At subsonic Mach, the pre-pass returns freestream conditions with no additional computation. Each downstream component calculator reads its local post-shock conditions from the pre-pass result rather than from freestream, correcting fin normal force slopes, dynamic pressure, interference factors, and drag coefficients.

All Mach regime transitions are C1-continuous. The compressibility factor $\beta = \sqrt{|M^2 - 1|}$ uses a cubic Hermite spline through M 0.95–1.05; base drag blends across M 0.85–1.3; Chapman-Korst turbulent base drag activates M 1.2–1.4; fin wave drag ramps in M 0.9–1.2; Modified Newtonian blends into the shock-expansion result M 4.0–6.0. The requirement for C1-continuity throughout is architectural: discontinuities cause the Runge-Kutta trajectory integrator to oscillate when the vehicle repeatedly crosses a Mach threshold.

An alternative `LookupTableDragCalculator` and `LookupTableStabilityCalculator` accept user-supplied CSV tables as overrides, enabling direct comparison against CFD or wind-tunnel data for specific vehicles.

Table 2 lists the C1-continuous blending regions that prevent trajectory integrator instabilities when the vehicle crosses Mach regime boundaries.

Table 2. Mach regime blending regions ensuring C1-continuous simulation.

Physical quantity	Blending range	Method
Compressibility factor β	M 0.95–1.05	Cubic Hermite spline
Skin friction	M 0.9–1.1	Polynomial blend
Base drag	M 0.85–1.3	C1 cubic blend
Fin wave drag	M 0.9–1.2	Hermite activation
Modified Newtonian	M 4.0–6.0	Smoothstep

Research Impact

Table 3 summarizes validation results for the principal subsystems.

Subsystem	Reference	MAPE
Oblique/normal shock relations	NACA 1135 (Ames Research Staff 1953)	< 0.01%
Nose wave drag (5 shapes)	NACA RM A52H28 (NACA Ames Aeronautical Laboratory 1952)	29.3%
Base drag, turbulent	NACA TN 3393 (Reller and Hamaker 1955)	15.9%
Base drag, laminar	NACA TN 3393 (Reller and Hamaker 1955)	4.4%
Fin wave drag	NACA TN 3650 (Ulmann 1956)	21%
Fin CNa (M 0.6–5.82)	NASA TM X-653 (Nielsen et al. 1962)	6.8%
Fin xCP (M 0.6–5.82)	NASA TM X-653 (Nielsen et al. 1962)	7.1%
Basic Finner total drag	ADA636861 (Dupuis and Hathaway 1997)	22.7%
Hypersonic cone drag (M 6.5–17.2)	DTIC AD0487365 (Grabow 1965)	17.8%

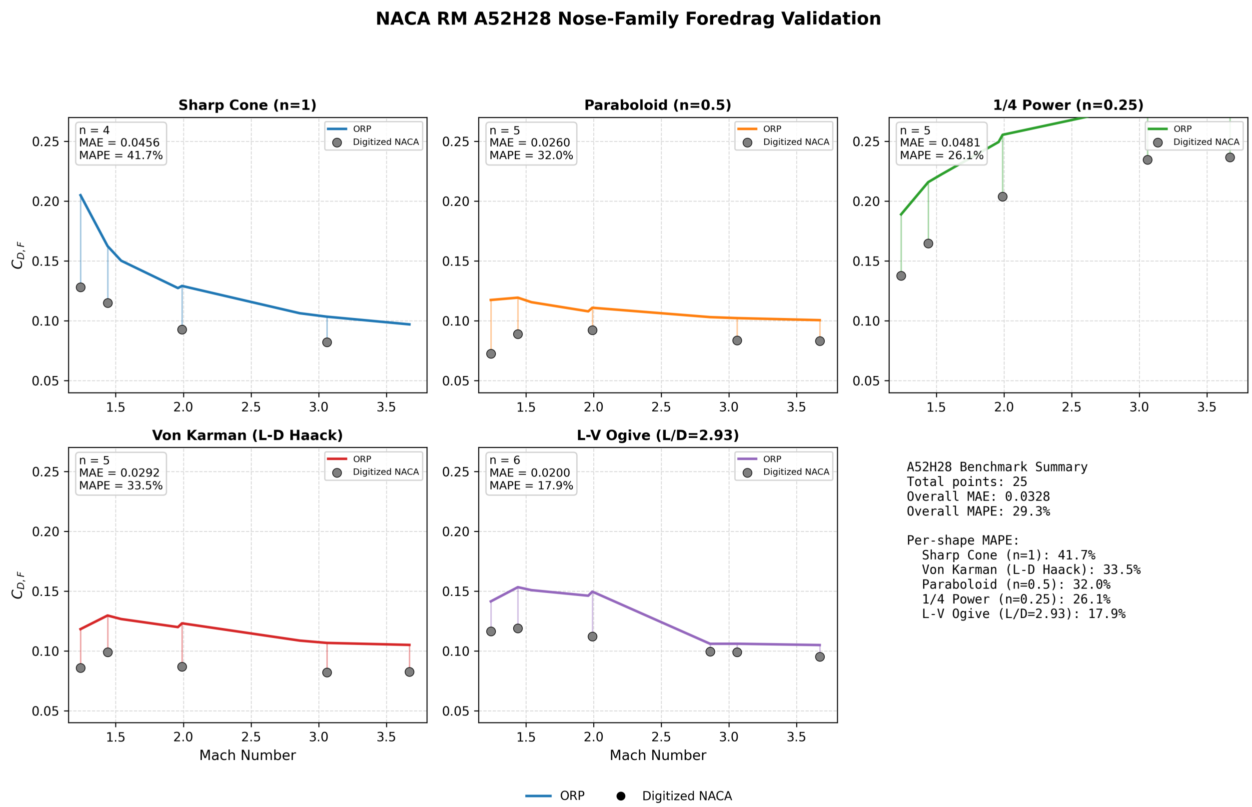

Figure 1: Nose wave drag coefficient versus Mach number for five nose shapes at fineness ratio 3, comparing the present model against NACA RM A52H28 wind-tunnel data. Aggregate MAE = 0.0328, MAPE = 29.3%.

The shock solver matches NACA 1135 tabular values to better than 0.01% across Mach 1.2–10 and cone half-angles 5–40°. Nose wave drag validated on five distinct nose shapes (MAPE 29.3%) covers the practical geometry space using Taylor-Maccoll (cones), second-order shock-expansion (ogives), and Dahlem-Buck shape factors (power-law and Haack Series). Fin aerodynamics validated against NASA TM X-653 — 26 data points spanning M 0.6 to 5.82 across multiple fin geometries — produce CNa MAPE 6.8% and center-of-pressure MAPE 7.1%. The Van Driest II compressible skin friction transformation (Hopkins and Inouye 1971) reduces C_f by approximately 50% relative to the incompressible value at Mach 5, consistent with the Hopkins and Inouye (1971) experimental survey.

The vehicle-level Basic Finner benchmark (MAPE 22.7%, 8 points, M 1.08–4.30) reflects real free-flight range measurements on 30 mm projectiles (Dupuis and Hathaway 1997), a more stringent test than wind-tunnel pressure data because it includes manufacturing variation, sabot separation, base flow unsteadiness, and all other physical effects present in actual flight. The hypersonic cone drag benchmark (MAPE 17.8%, 11 points, M 6.5–17.2) confirms usable accuracy through the Modified Newtonian regime.

The software enables a class of research and educational activities that was previously inaccessible without commercial tools: trajectory optimization for HPR vehicles at Mach 2–5, aerodynamic stability analysis across the full transonic-to-supersonic transition, and open, reproducible comparison of competing aerodynamic models. The extensible architecture — each model is an independently testable Java class with explicit Mach validity ranges — makes it straightforward to incorporate improved correlations as the research literature advances.

Known limitations include elevated MAPE for nose shapes where small absolute drag values amplify percentage errors (power-law and Haack noses via empirical Dahlem-Buck shape factors), systematic underprediction of total vehicle drag on the Basic Finner projectile suggesting a missing interference or transition drag source, and reduced center-of-pressure accuracy (xCP MAPE 7.1%) that degrades in the transonic band where the transonic similarity approximation is least accurate. The transonic regime (M 0.8–1.3) remains the most challenging for all models. Real-gas dissociation chemistry above approximately Mach 7 is not included.

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